

Ethanol Blending in Gasoline – Austria

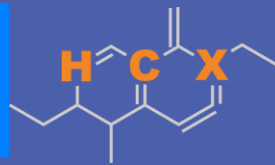
September 2025



**U.S. GRAINS &
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20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90



www.faro90.com
+52 55 6122 2255

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Austria

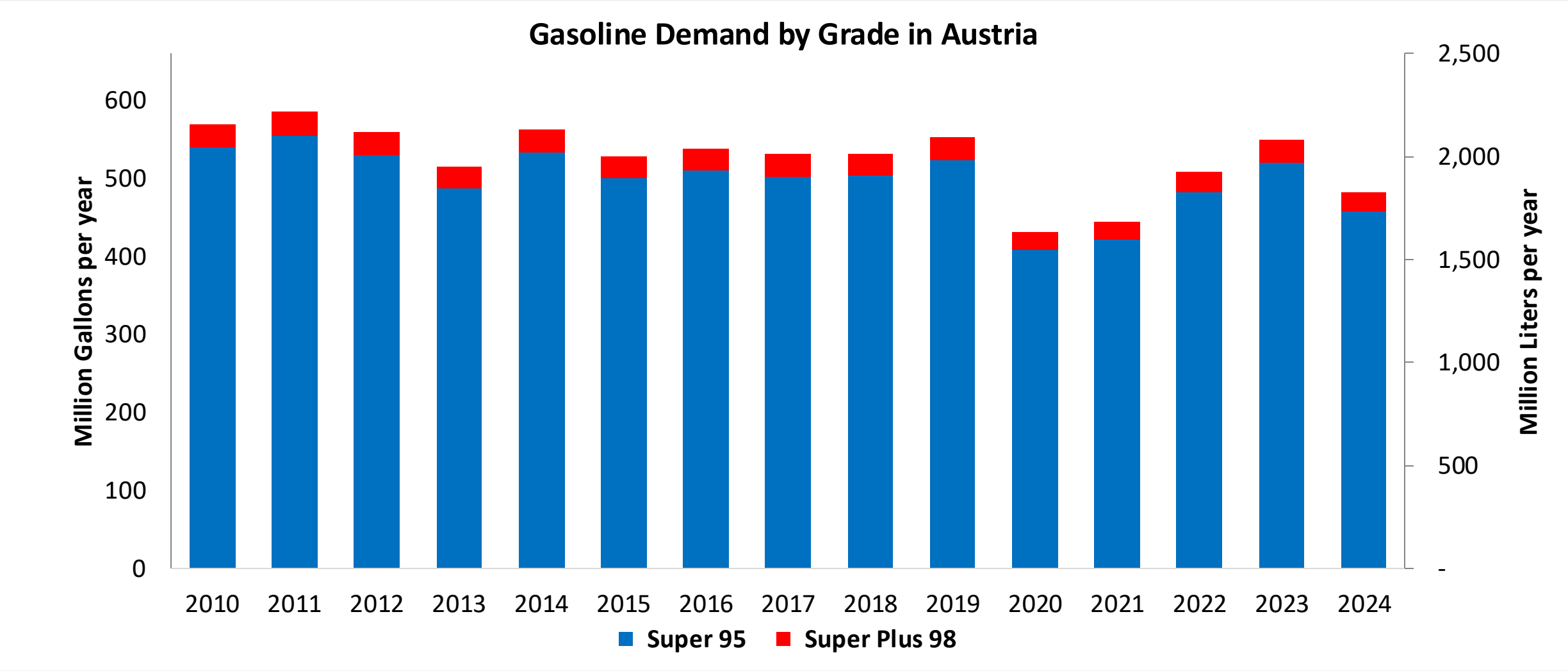


In 2024, gasoline consumption in Austria was 1,829 million liters (483 million gallons) and growth rate was -2.7%. Gasoline production and net imports were 2,317 and -392 million liters (612 and -103 million gallons) correspondingly. Austria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Austria was 185 million liters (49 million gallons) representing 10.1% of gasoline demand. Ethanol production and Net Imports were 297 and -112 million liters (49 and -30 million gallons) correspondingly. Ethanol sources are Corn 75%, Cereals 15%, and Sugar Beet 10%.

Source: eurostat

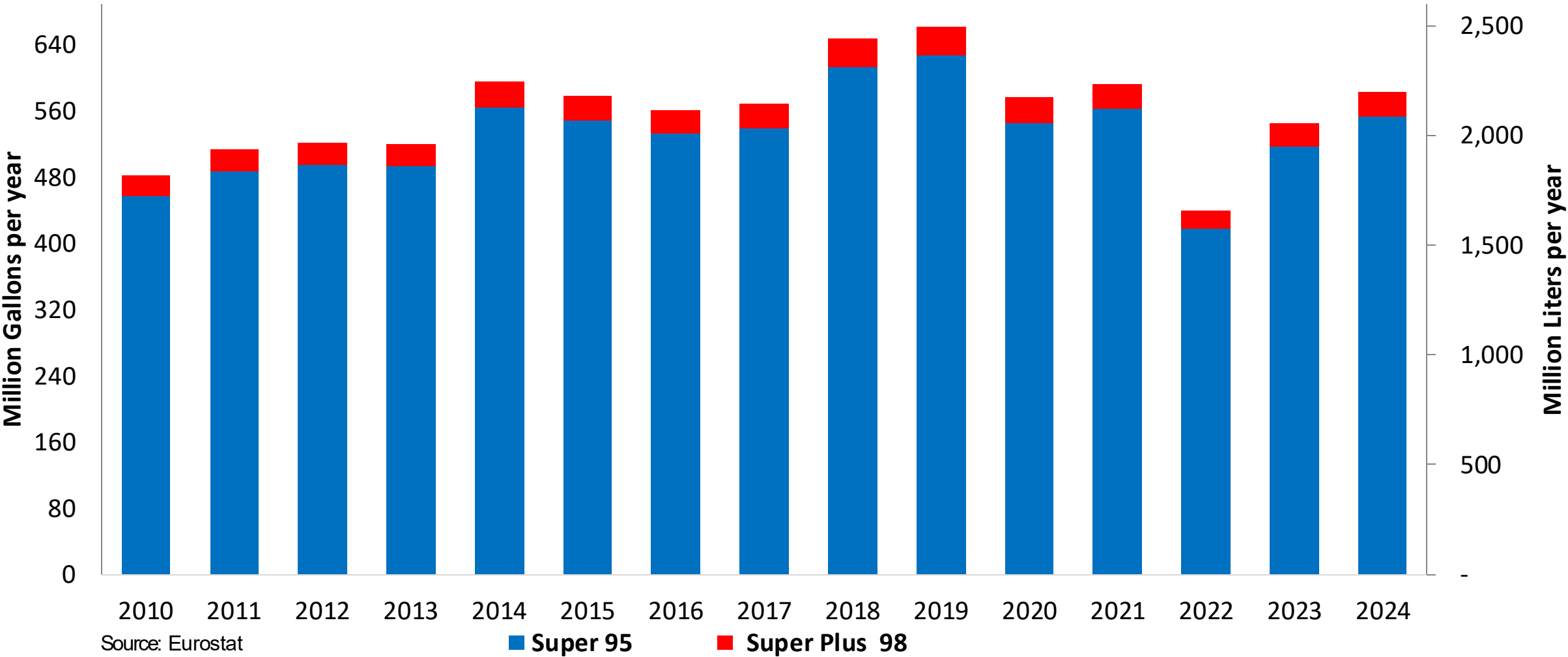
Gasoline Demand in Austria



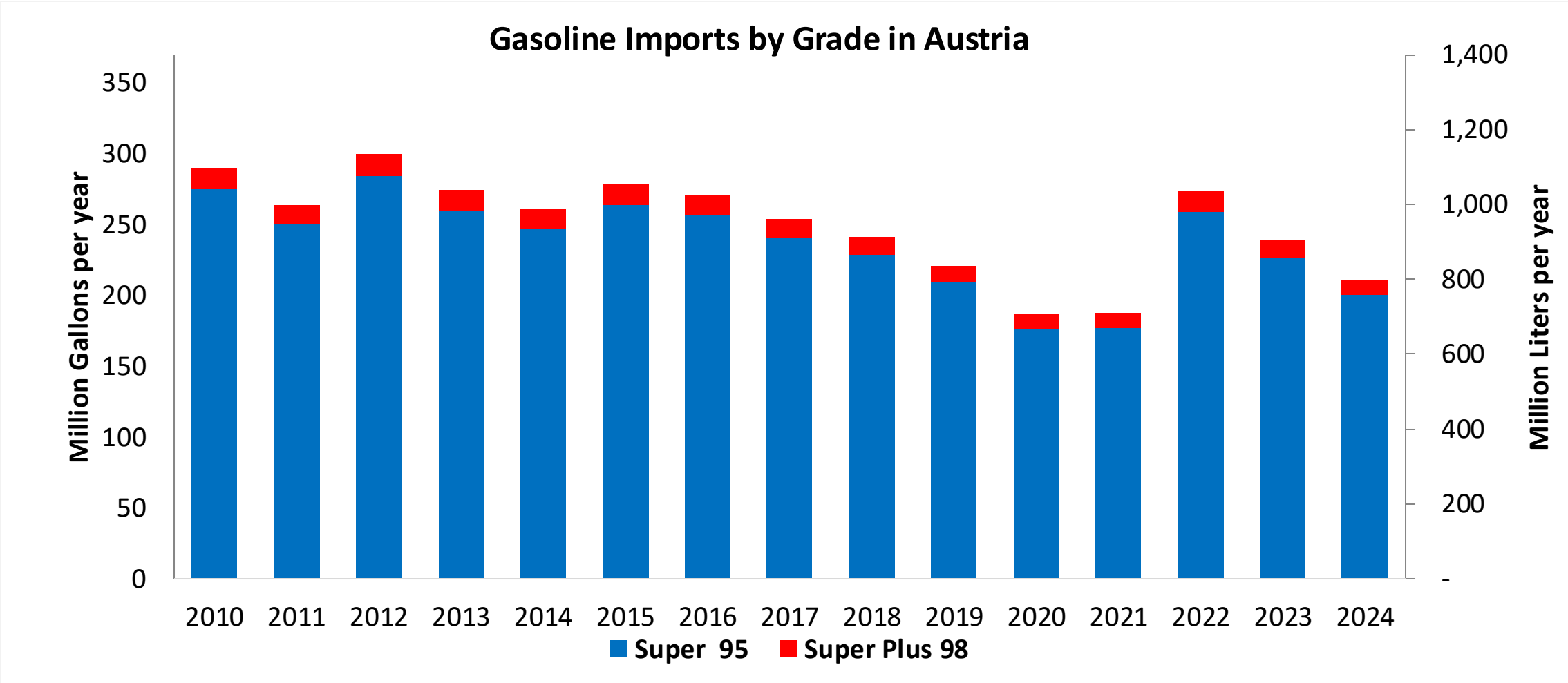
Source: Eurostat

Gasoline Production in Austria

Gasoline Production by Grade in Austria

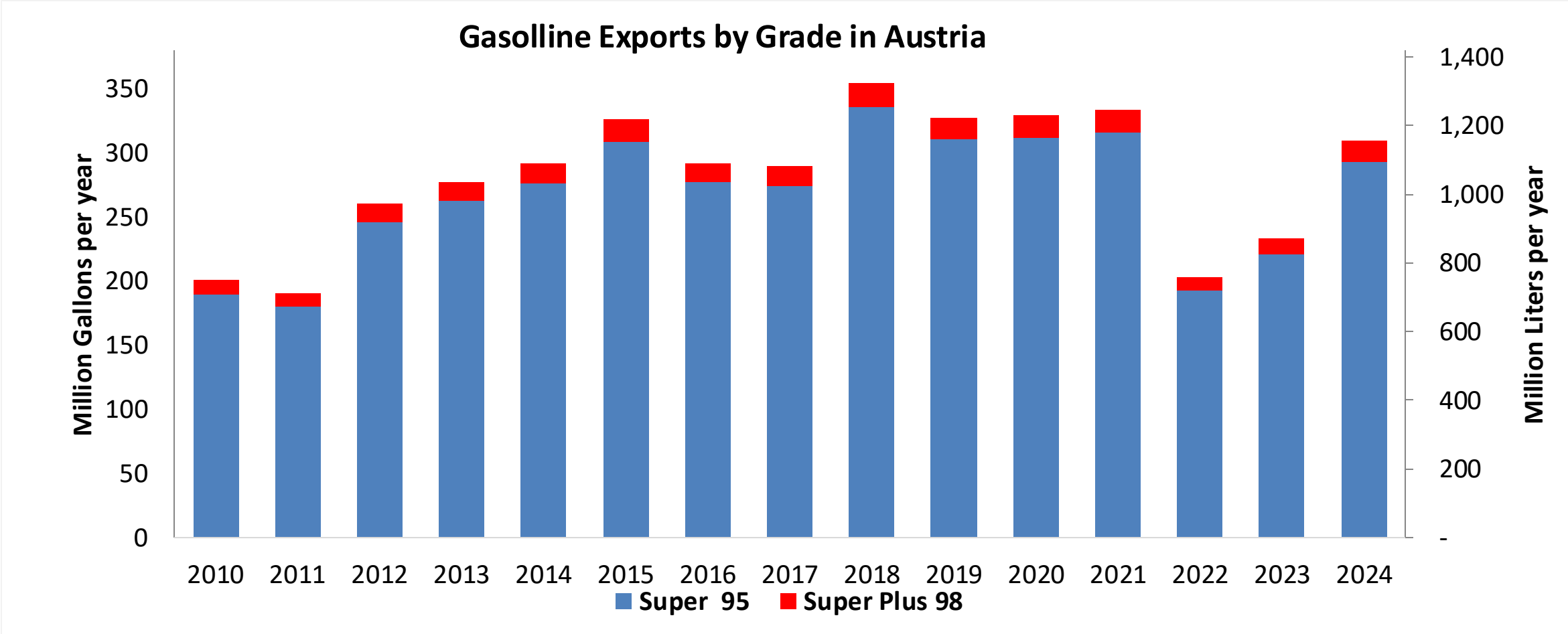


Gasoline Imports in Austria



Source: Eurostat

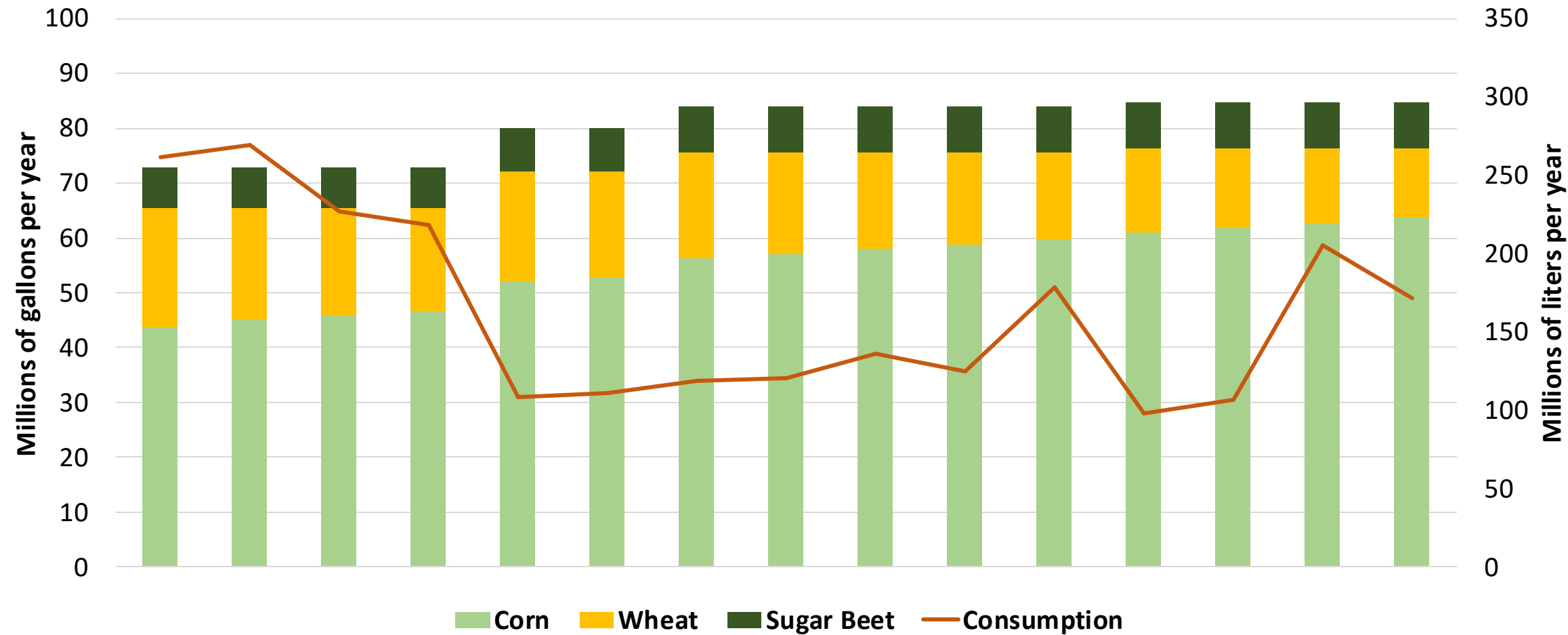
Gasoline Exports in Austria



Source: Eurostat

Ethanol Balance in Austria

Ethanol Balance in Austria



Source: Eurostat

Gasoline Quality in Austria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	3.4							
GHG Emission Reduction (%)	7.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

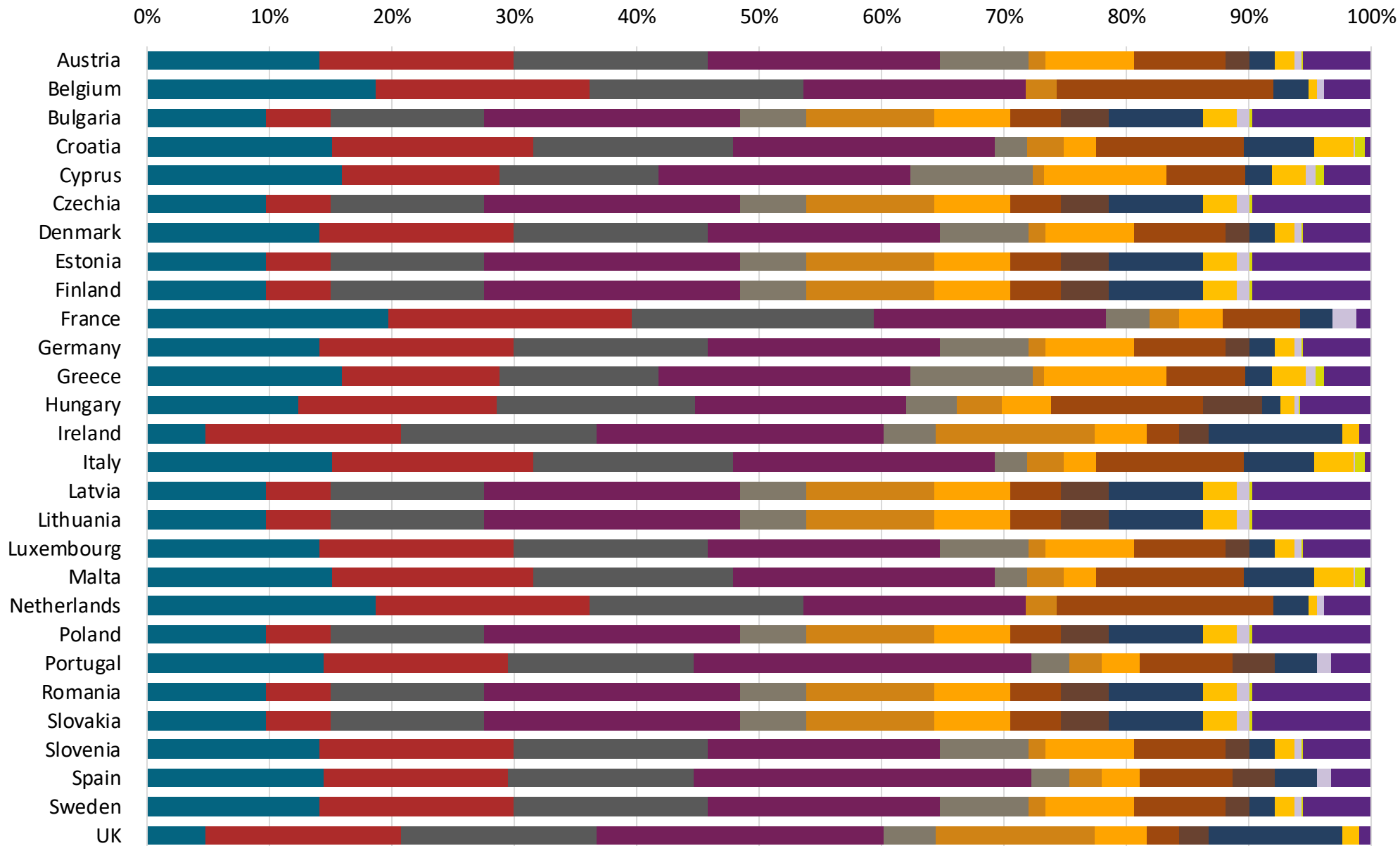
Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90



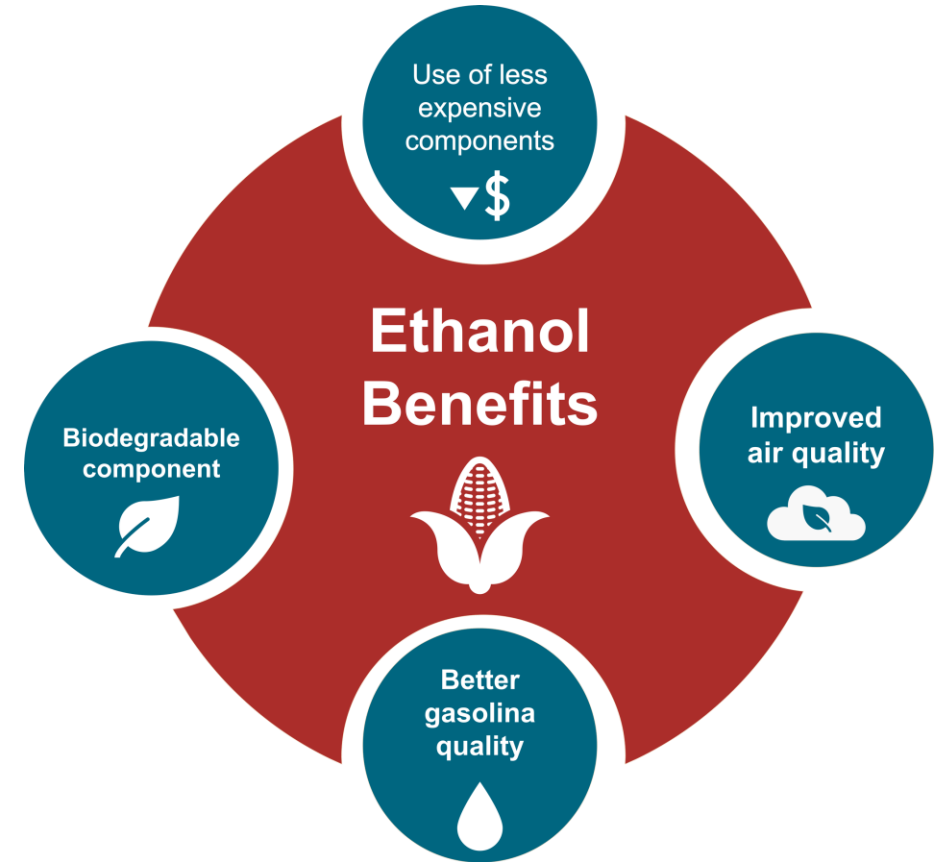
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

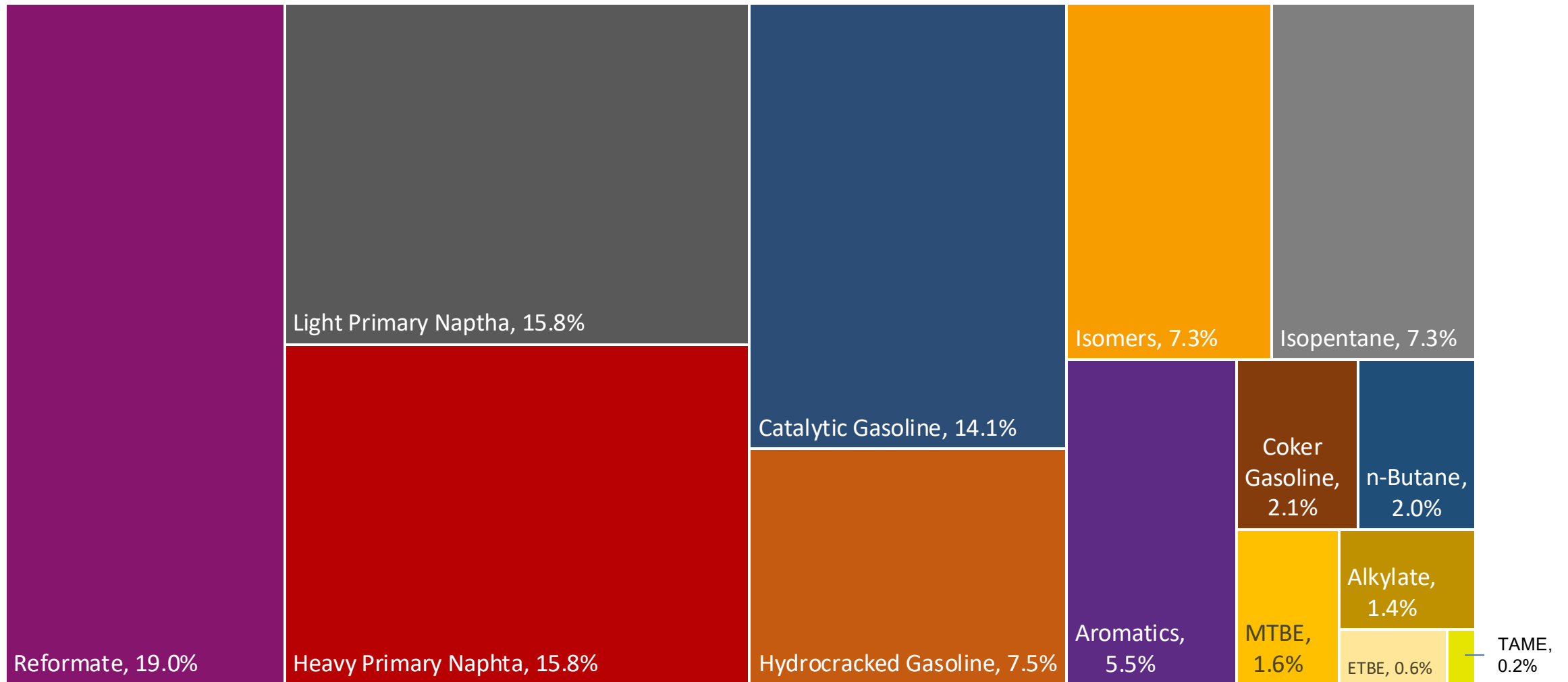
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

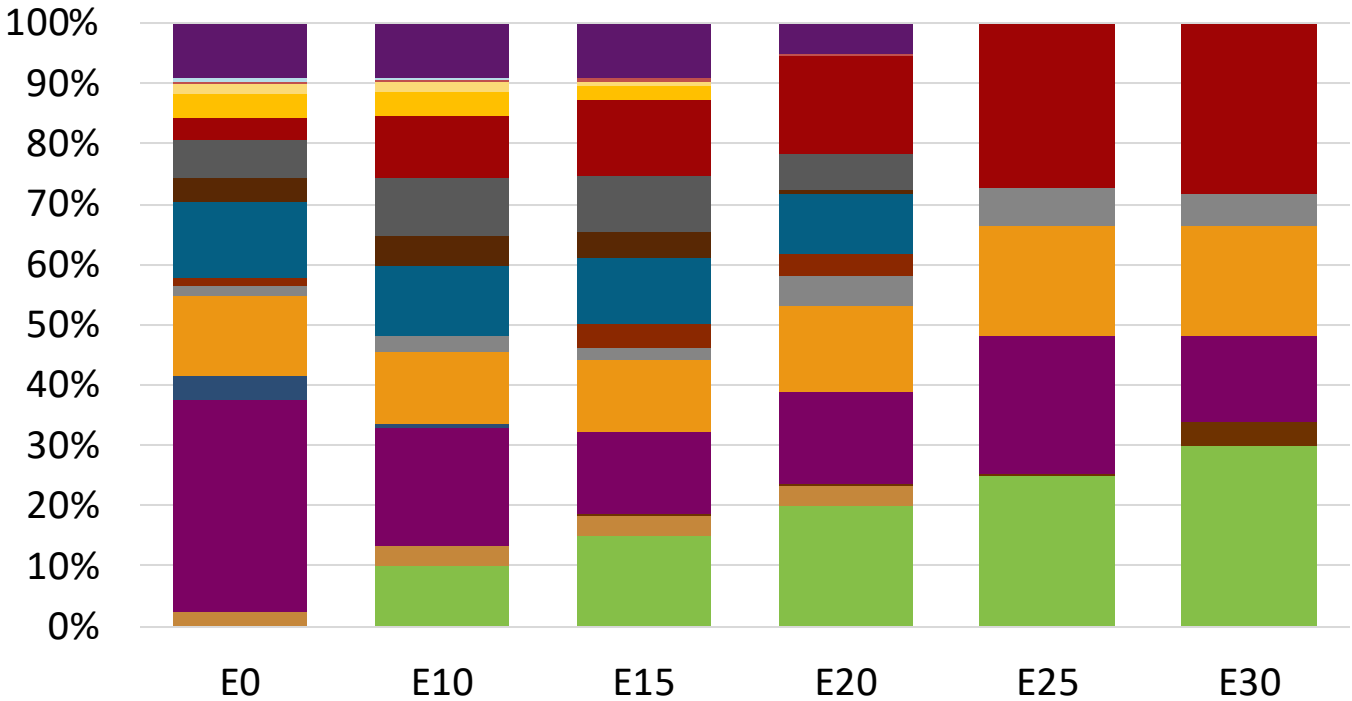
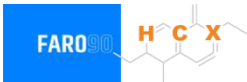
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Austria Available Blending Components



Austria – Gasoline 95 – Constant Octane

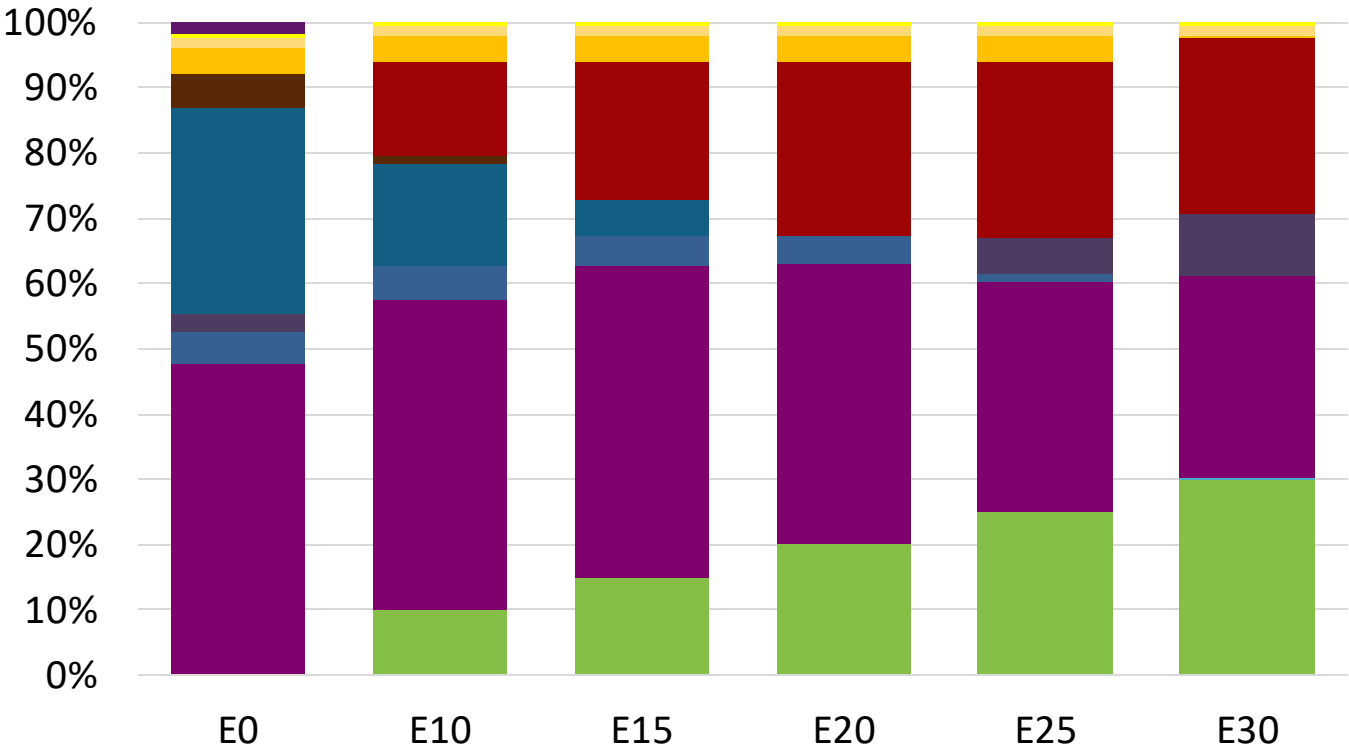


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

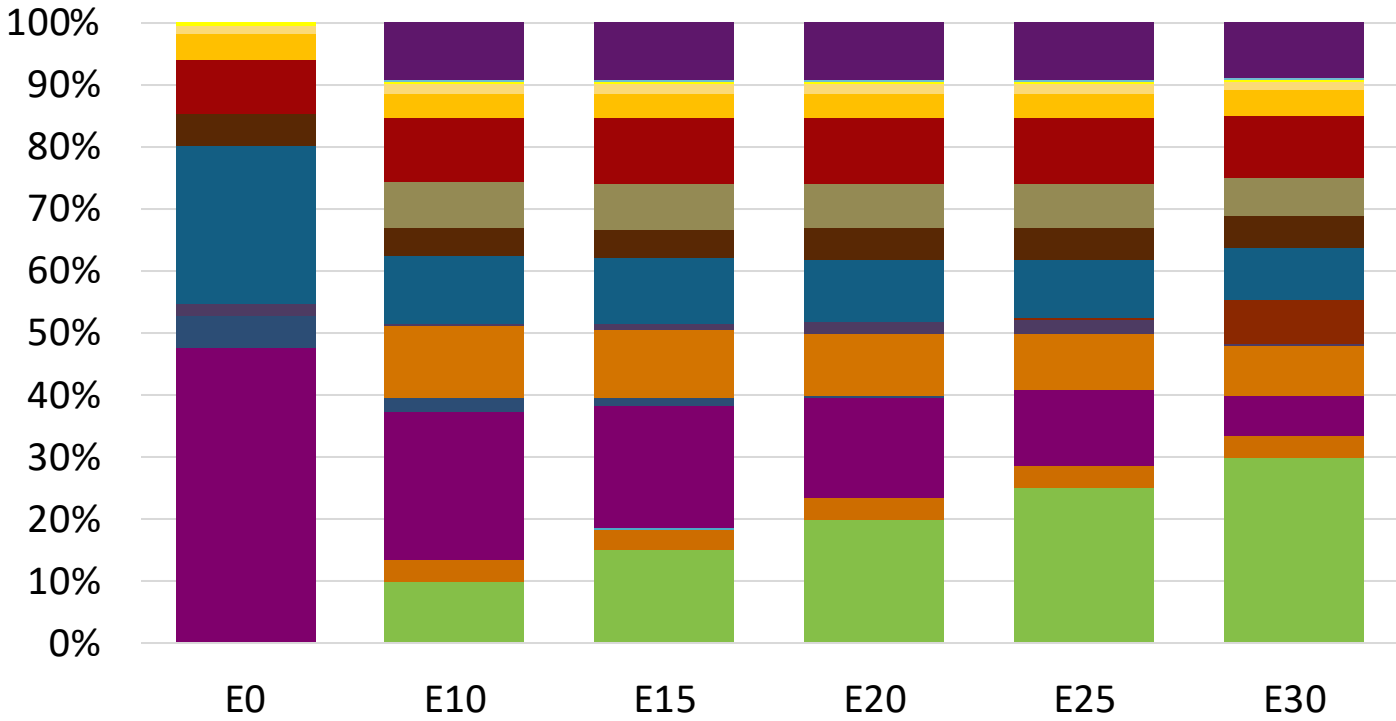
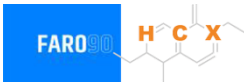
Austria - Gasoline 98 - Constant Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

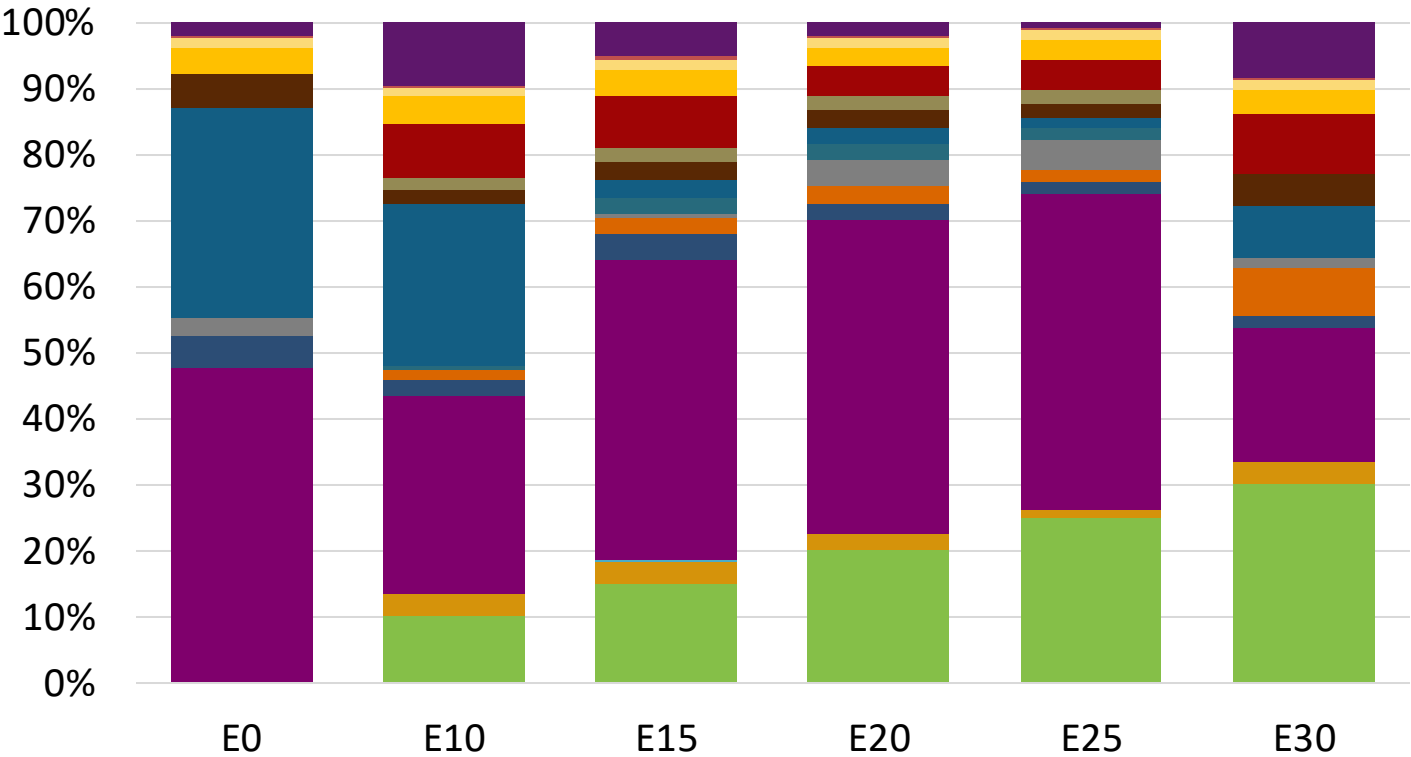
Austria – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Austria – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)		98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$	2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

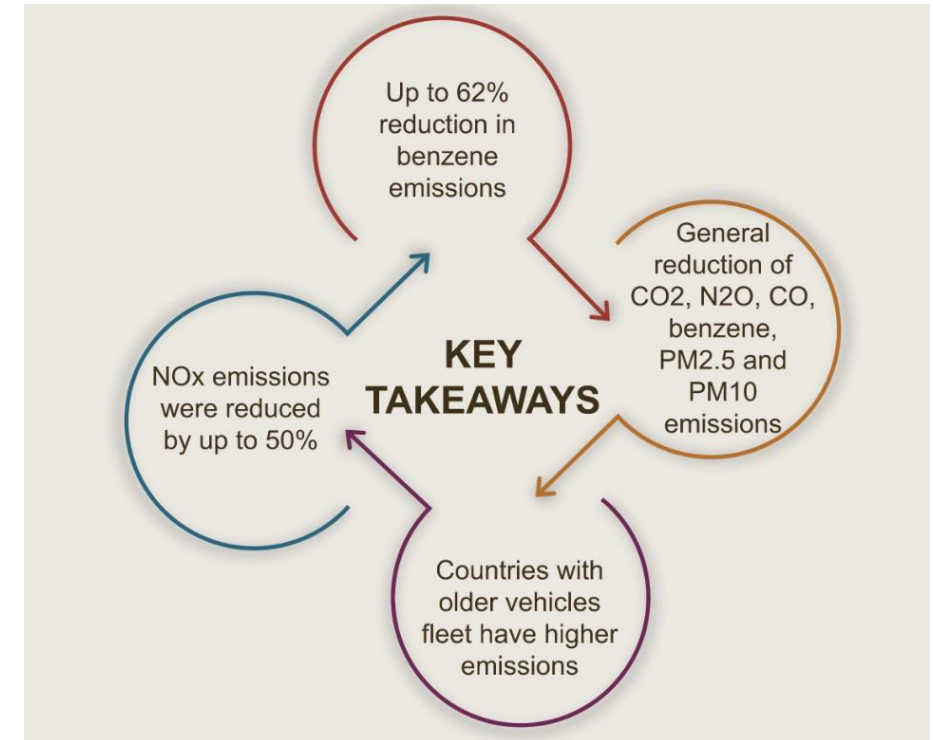
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

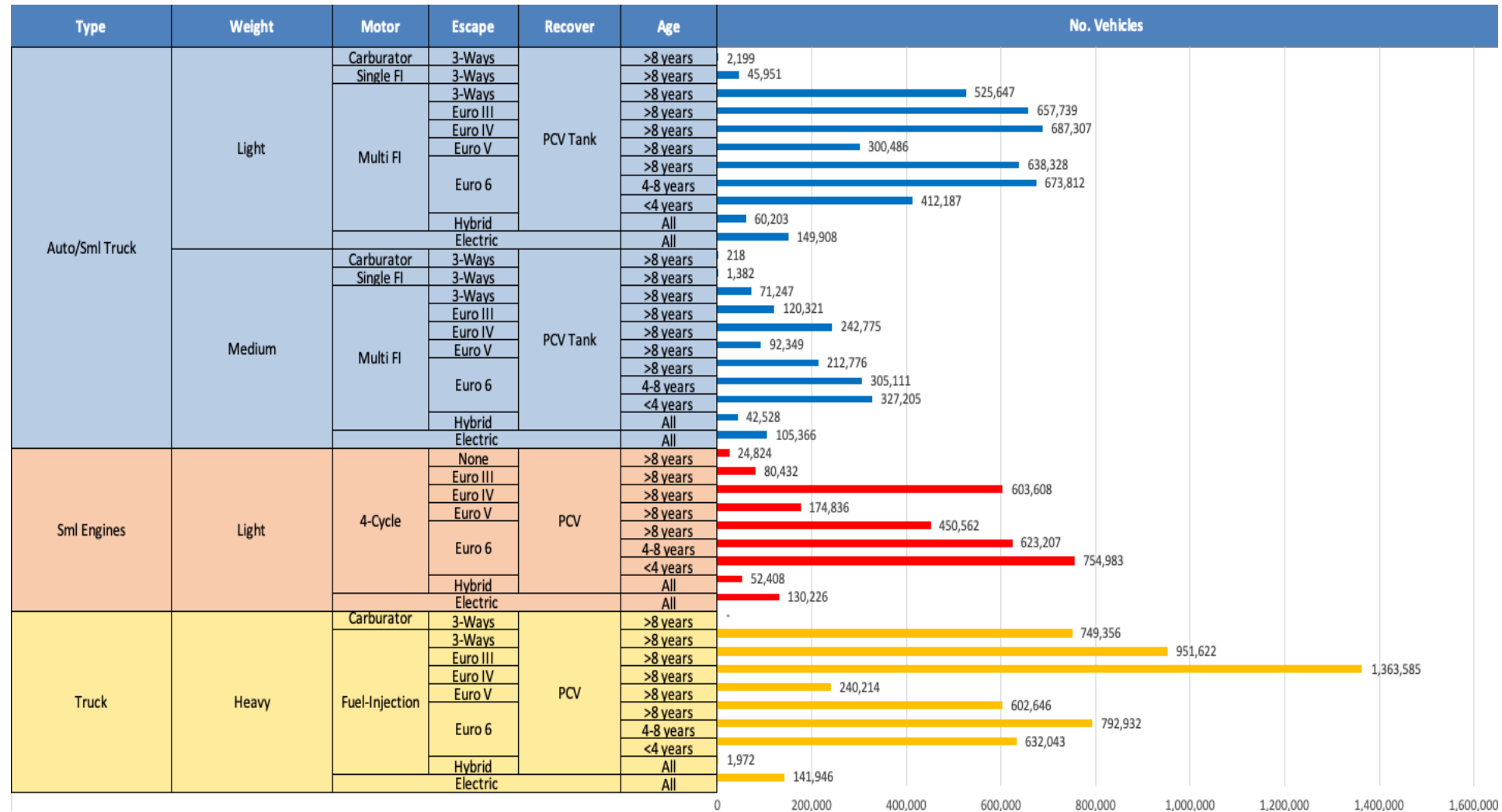
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Austria

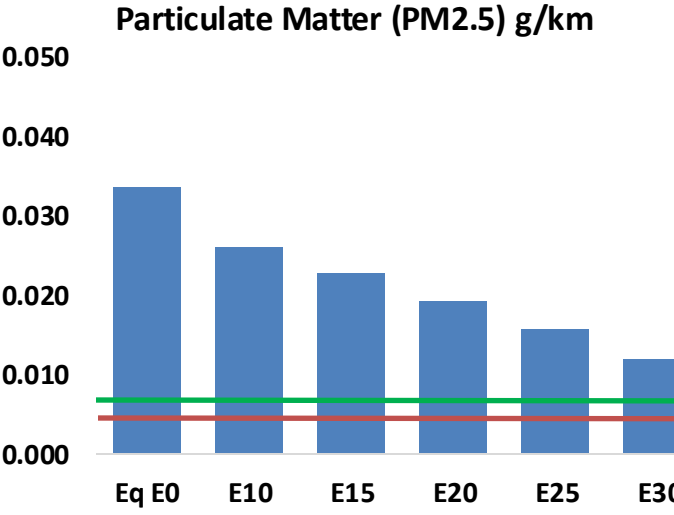
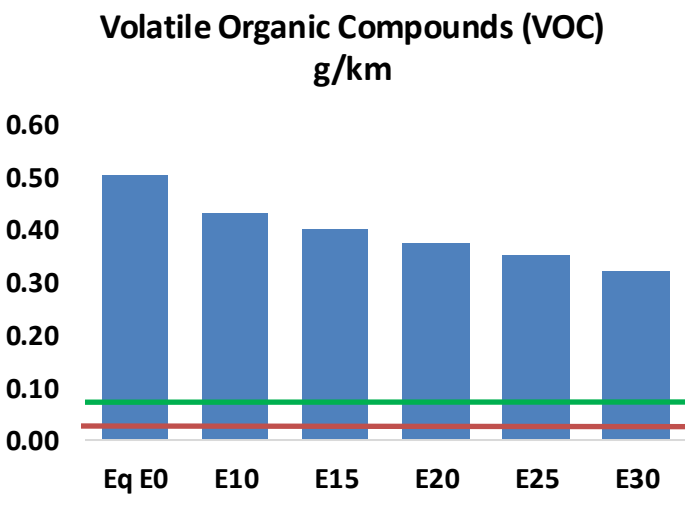
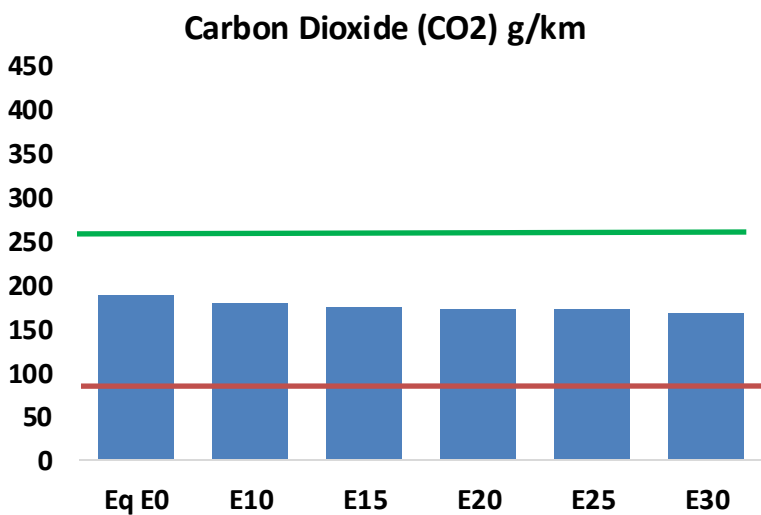
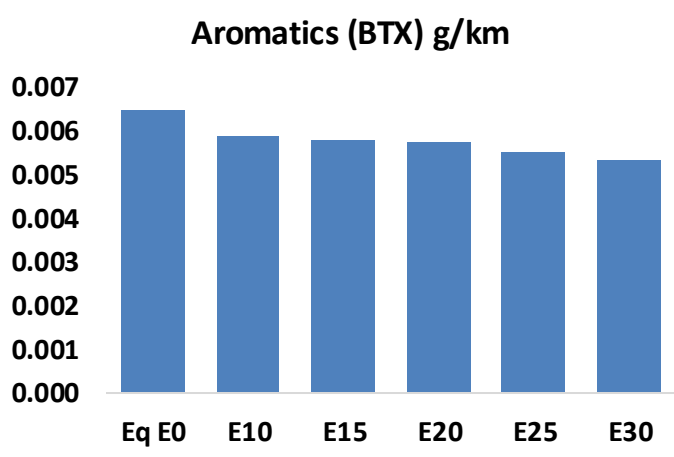
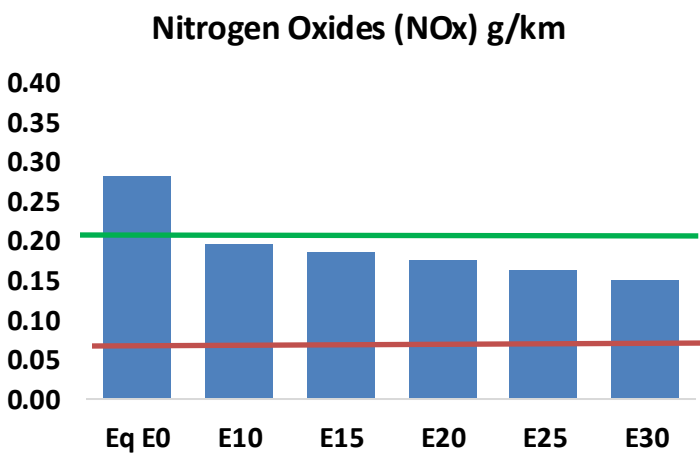
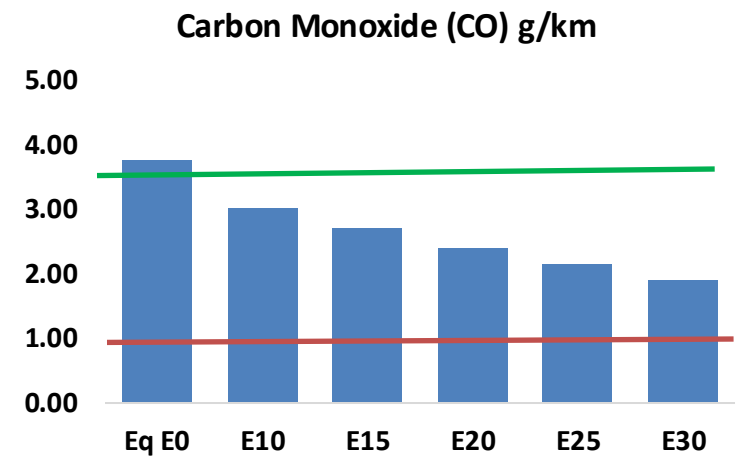
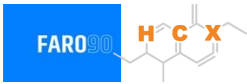


Vehicle Fleet: 14,046,44
Gasoline Vehicle Fleet: 3,251,450

Source: Eurostat, ACEA

Austria – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Austria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	315,891	285,353	254,815	228,570	201,597	180,687	158,890	-19%	-36%
CO2	15,777,119	15,371,421	14,988,263	14,686,920	14,537,804	14,452,599	14,186,196	-5%	-8%
VOC	42,267	39,233	36,199	33,755	31,314	29,444	26,927	-14%	-26%
NOx	23,733	17,800	16,613	15,658	14,767	13,784	12,745	-30%	-38%
SOx	178	165	151	140	129	117	105	-15%	-28%
NH3	5,682	5,626	5,569	5,596	5,659	5,703	5,740	-2%	0%
N2O	251	249	248	250	255	258	261	-1%	2%
CH4	9,822	9,822	9,822	9,969	10,186	10,372	10,549	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	313	297	281	270	259	251	239	-10%	-17%
Acetaldehyde	819	940	1,379	2,100	2,861	3,270	3,863	68%	249%
Formaldehyde	2,974	2,892	3,371	3,958	4,146	4,587	4,994	13%	39%
Benzene	545	523	496	488	484	463	448	-9%	-11%
PM2.5	1,014	900	787	683	580	471	357	-22%	-43%
PM10	1,813	1,610	1,407	1,222	1,037	842	638	-22%	-43%

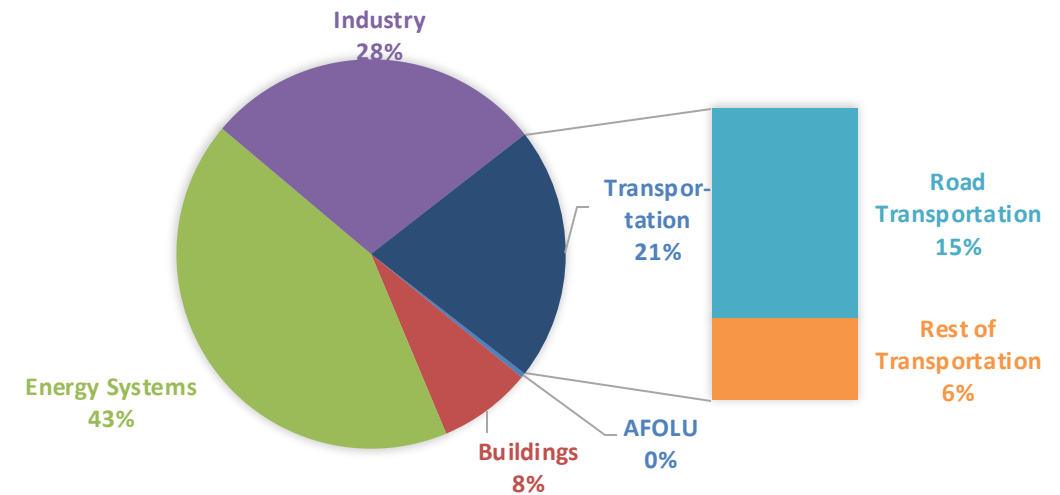
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

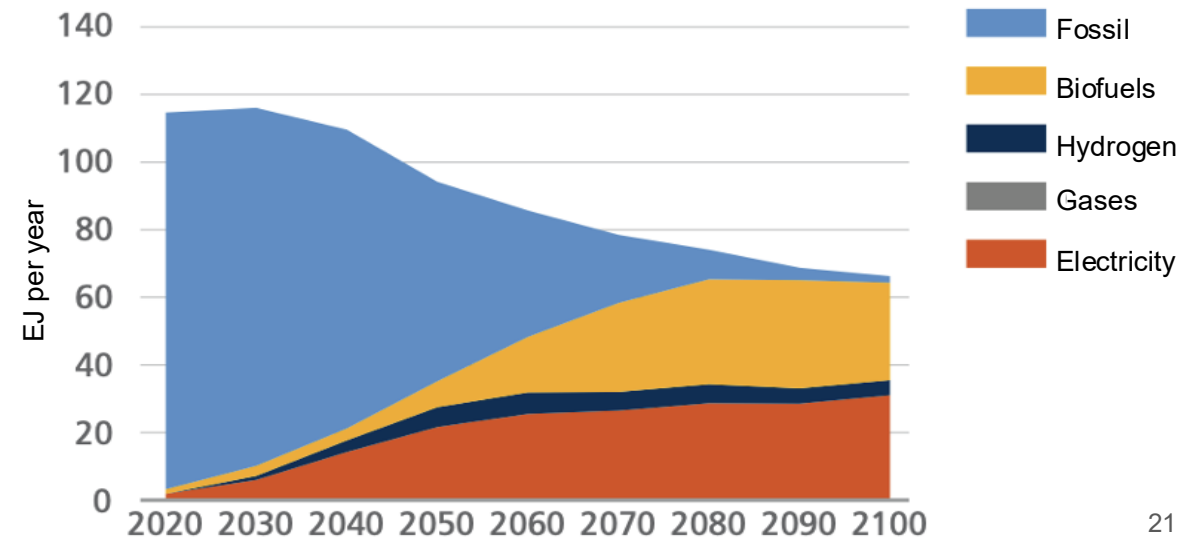
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

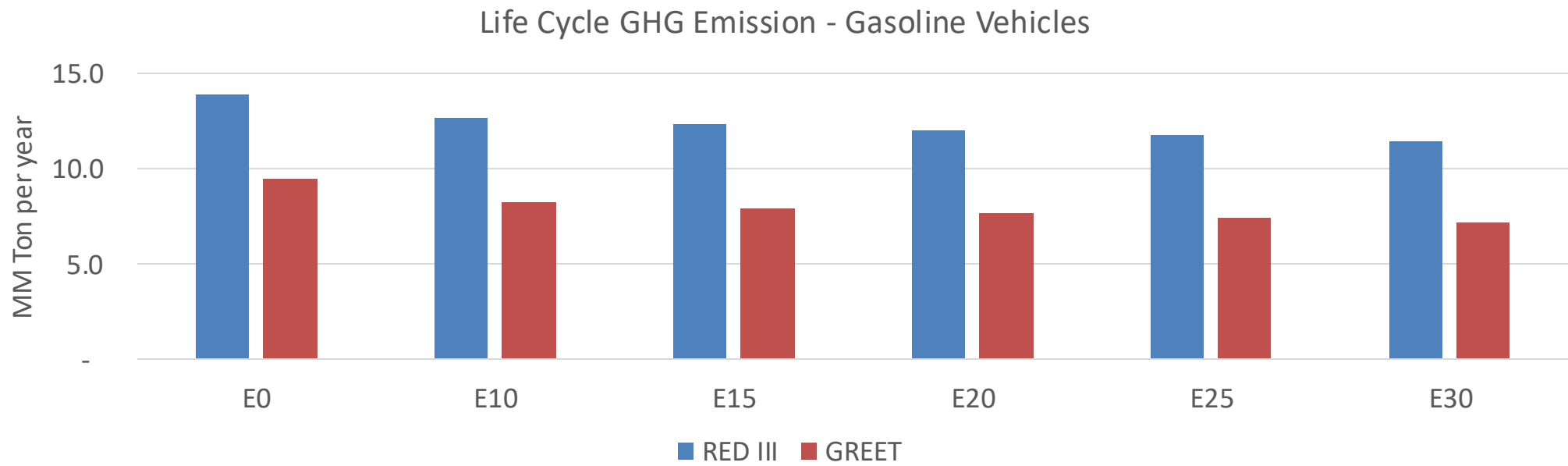
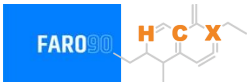
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Austria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	67.5
Target @ 2035 (MMT/year)	38.0
Delta	29.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.9%	16.1%	18.6%	21.0%	24.2%
RED II/III	0.0%	9.3%	11.7%	13.6%	15.4%	17.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.1%	5.1%	6.0%	6.7%	7.7%
RED II/III	0.0%	4.4%	5.5%	6.4%	7.3%	8.4%

Source: Greet, RED II/III, climateactiontracker.org, F90